CIVICSIFT AI

WEEK 1 – NLP PROJECT PLANNING AND PROPOSAL REPORT

An Evidence-Aware Multilingual NLP Framework for Citizen Complaint Triage and Incident Discovery

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PROJECT PLANNING REPORT

Report Overview

This report presents the planning and proposed design of CivicSift AI, a multilingual Natural Language Processing (NLP) application for organising citizen complaints. It covers the problem context, project objectives, research background, proposed methods, tools, evaluation strategy, implementation schedule, risks and expected outcomes. The work described here is a project plan; it does not claim that a model has already been trained or that experimental results have been achieved.

1. Introduction

Public-service departments may receive reports about damaged roads, water interruptions, overflowing waste, drainage problems, electricity faults and public-space hazards. Such reports are commonly written as short free-text messages. People may describe the same issue in different ways, use local language expressions, switch between languages, or type regional-language words using the Roman alphabet. These variations make manual sorting and keyword-based processing difficult.

Natural Language Processing offers techniques for analysing text, identifying topics, extracting information and measuring semantic similarity. CivicSift AI is proposed as a human-assisted system that converts incoming complaint text into structured information so that a reviewer can understand and organise reports more efficiently.

2. Project Title and Summary

Project title: CivicSift AI – An Evidence-Aware Multilingual NLP Framework for Citizen Complaint Triage and Incident Discovery.

The proposed system will accept complaint text and suggest a service category, an urgency band, relevant information mentions and semantically similar complaints. Its initial language scope is English, Hindi and Kannada, with a small controlled evaluation of Romanised and code-mixed text. The system will display its confidence and supporting text cues and will leave final decisions to a human reviewer.

3. Problem Statement

Citizen complaints are unstructured, multilingual and sometimes repetitive. Manual triage can lead to inconsistent categorisation and make it difficult to recognise multiple reports that may describe the same underlying incident. Basic keyword matching may fail when messages use different vocabulary or languages. There is therefore a need to investigate a practical NLP workflow that can classify complaints, surface potential urgency indicators and retrieve related reports while making uncertainty visible.

4. Aim and Objectives

Aim: To design and evaluate a lightweight multilingual NLP prototype that supports the classification, prioritisation and consolidation of citizen complaints.

1. Define a complaint taxonomy for common public-service issues such as water, roads, sanitation, drainage, electricity and safety hazards.
2. Prepare a privacy-conscious dataset of representative English, Hindi, Kannada and code-mixed complaint examples.
3. Build a TF-IDF and classical machine-learning baseline for complaint category classification.
4. Compare the baseline with a multilingual transformer model such as IndicBERT or XLM-R.

5. Estimate urgency using labelled examples and transparent text cues, with low-confidence cases referred for human review.
6. Retrieve potential duplicate complaints using multilingual sentence embeddings and semantic similarity.
7. Evaluate model performance using macro-F1, per-language analysis, duplicate-retrieval metrics and error analysis.
8. Develop a simple interface that displays predictions, confidence, supporting phrases and related complaints.

5. Need for an NLP-Based Solution

- Language diversity: A complaint may be expressed in English, Hindi, Kannada, transliteration or a mixture of languages.
- Meaning-based matching: Semantic representations can identify related messages even when they do not share exact keywords.
- Consistent first-pass organisation: Classification can provide a repeatable initial category suggestion for human review.
- Incident awareness: Similarity search can bring potentially related reports together for inspection.
- Transparent prioritisation: Highlighting urgency-related phrases makes a suggested priority easier to review than an unexplained score.

NLP is suitable because the primary input is natural-language text. It will support, not replace, the judgement of authorised public-service personnel.

6. Literature Review and Research Gap

6.1 Indic-language NLP

Kakwani et al. (2020) introduced IndicNLPSuite, bringing together monolingual corpora, evaluation benchmarks and pretrained models for Indian languages. This work provides a foundation for studying language understanding beyond English and motivates the use of Indic-focused resources in this project.

6.2 Multilingual language understanding

IndicBERT and related AI4Bharat resources provide pretrained representations for Indian languages. XLM-R is another multilingual transformer that can be considered as a comparison model. These models offer a practical alternative to training a large language model from scratch, although their performance on local, informal complaint text must be measured rather than assumed.

6.3 Code-mixed language

Code-mixed text presents additional challenges because a single message may contain multiple languages, scripts and informal spellings. COMI-LINGUA (Findings of EMNLP 2025) is an example of research that develops expert-annotated Hindi-English code-mixed resources for NLP tasks. Its relevance is that code-mixed data should be included deliberately in training and evaluation rather than treated as ordinary monolingual text.

6.4 Research opportunity

Civic complaint classification and Indic-language NLP have existing research foundations; therefore, this proposal does not claim that multilingual complaint classification is entirely new. The proposed

contribution is to investigate a combined, human-in-the-loop workflow that links multilingual category prediction with semantic duplicate discovery and evidence-aware urgency suggestions, while explicitly evaluating language-specific errors and uncertainty.

The literature review will be expanded during implementation by checking recent papers, dataset documentation, licences and model cards. Any claimed research gap will be refined after this review.

7. Proposed Methodology

7.1 Data preparation

The project will use public datasets only where their licences permit the intended use, along with clearly labelled synthetic or consented examples. A proof-of-concept target of approximately 1,000–2,000 examples is proposed, subject to data availability and annotation capacity. Each record may include complaint text, language/script, category, urgency label, duplicate-group identifier and optional location mention.

- Prepare written annotation guidelines with examples for every category and urgency level.
- Remove or mask unnecessary personal identifiers such as phone numbers and personal names.
- Record each sample's source, licence, language and whether it is synthetic or real.
- Split data by incident or duplicate group to prevent near-identical messages appearing in both training and test sets.
- Review a subset with a second annotator where feasible and resolve disagreements.

7.2 Text preprocessing and language handling

The pipeline will apply Unicode normalisation and careful whitespace cleanup while preserving negation, quantities, dates and urgency-bearing terms. Language and script will be estimated where feasible. Original text will be retained for authorised review, and any transliteration or spelling normalisation will be documented so that useful meaning is not accidentally removed.

7.3 Complaint category classification

A baseline model will use word- and character-level TF-IDF features with Logistic Regression or Linear Support Vector Machine. The advanced approach will evaluate a pretrained multilingual transformer such as IndicBERT or XLM-R using a classification head. Both approaches will use the same data splits. The more complex model will be preferred only if it provides a meaningful improvement relative to its compute and maintenance cost.

7.4 Urgency estimation

Urgency will be modelled separately from complaint category. A rule-based baseline will identify explicit risk phrases, and a supervised classifier may be tested if enough consistently labelled examples are available. The system will display the evidence phrases behind a suggestion and provide an insufficient-information option. Sentiment alone will not be treated as a reliable measure of urgency.

7.5 Duplicate complaint discovery

A multilingual sentence-embedding model will represent complaint messages as vectors. Cosine similarity will retrieve candidate related reports. Thresholds will be selected using labelled validation pairs. Clustering methods such as agglomerative clustering or DBSCAN may be explored if the dataset

is large enough. Similarity will indicate a candidate match, not prove that two reports describe the same event; a reviewer must confirm any grouping.

7.6 Information extraction and interface

The system may extract explicit references to locations, dates, services and infrastructure using rules or suitable NLP libraries. These are text-derived mentions and will not be treated as verified geographic facts. A Streamlit interface will show the complaint, suggested category, urgency band, confidence, evidence phrases and similar reports, with an option for a reviewer to correct predictions.

8. Proposed System Workflow

Step	Operation	Output
1. Input	Receive an authorised complaint message	Raw complaint text
2. Privacy and preparation	Mask unnecessary identifiers and normalise text	Prepared text
3. Language analysis	Estimate language, script and code-mixing	Language metadata
4. NLP analysis	Predict category, estimate urgency and extract mentions	Structured complaint
5. Similarity search	Compare multilingual sentence embeddings	Related-report candidates
6. Human review	Inspect evidence, confidence and candidates	Confirmed or corrected record
7. Evaluation	Measure model performance and review errors	Evaluation report

9. Tools and Technologies

Area	Proposed tools
Programming	Python
Data processing	Pandas, NumPy
Text processing	Regular expressions, NLTK or spaCy; Indic-language resources where suitable
Classical machine learning	Scikit-learn: TF-IDF, Logistic Regression, Linear SVM
Transformer models	Hugging Face Transformers; IndicBERT or XLM-R
Sentence similarity	Sentence-Transformers, cosine similarity
Interface	Streamlit
Development	Jupyter Notebook / Google Colab, Git and GitHub
Evaluation	Scikit-learn metrics, confusion matrices and manual error analysis

10. Evaluation Strategy

A held-out test set will be separated by incident group. Results will be reported overall and separately by language or script when sample sizes allow. Baseline and transformer models will be compared under the same experimental conditions.

Component	Evaluation measures	Provisional target*
Category classification	Macro-F1, per-class precision/recall, confusion matrix	Macro-F1 of 0.75 or higher
Urgency estimation	Macro-F1, recall for critical examples, confidence review	Critical-class recall of 0.80 where data permits
Duplicate retrieval	Precision@5, Recall@5, pairwise F1 or cluster purity	Precision@5 of 0.75 or higher
Language robustness	Per-language macro-F1, performance gaps, error analysis	Report gaps and limitations transparently
Prototype	Functional tests, response time, reviewer task checks	Successful end-to-end demonstration

*These are proposed proof-of-concept targets, not achieved results or guarantees. They will be revised after a pilot dataset and baseline evaluation.

11. Six-Week Implementation Timeline

Week	Planned activities	Deliverable
1	Literature review, project scope, complaint taxonomy, privacy and data-source review	Approved proposal and plan
2	Dataset collection, pilot annotation, class-balance analysis	Version 1 dataset and annotation guide
3	Preprocessing, language handling and TF-IDF baseline	Baseline notebook and metrics
4	Multilingual transformer evaluation and per-language error analysis	Model comparison report
5	Urgency cues, urgency model and semantic duplicate retrieval	Triage and similarity modules
6	Streamlit integration, testing, final evaluation and documentation	Working prototype and final report

The schedule assumes timely access to data and computing resources. If model fine-tuning is not feasible, the fallback is a reproducible classical baseline, embedding-based duplicate retrieval and a transparent rule-assisted urgency module.

12. Risk Analysis and Mitigation

Risk	Impact	Mitigation
Limited labelled data	Unstable performance	Pilot annotation; permitted public and synthetic data; disclose limitations.
Uneven language representation	Poorer results for underrepresented languages	Track data balance and report per-language metrics.
Code-mixed spelling variation	Language detection and classification errors	Include Romanised examples; test character n-grams and documented normalisation.
Incorrect urgency prediction	False escalation or missed risk	Human review, evidence display, conservative thresholds and abstention.
False duplicate match	Unrelated reports grouped together	Tune retrieval thresholds and require reviewer confirmation.
Privacy exposure	Disclosure of personal information	Data minimisation, masking, access restriction and synthetic demo data.
Compute limitations	Delayed transformer experiments	Use pretrained models, small batches, Colab and classical fallback.
Dataset/model licence restrictions	Use may not be permitted	Check licences and terms before use; document provenance.

13. Ethical and Privacy Considerations

- The prototype is a decision-support tool; authorised humans retain responsibility for final decisions.
- Only public, licensed, synthetic or explicitly consented data will be used.
- Personal identifiers will be removed or masked unless there is a documented and authorised need.
- Sensitive personal characteristics will not be used to rank or prioritise complaints.
- Language style, grammar or sentiment will not be used to judge a complainant's credibility.
- Low-confidence or ambiguous cases will be sent for manual review rather than forced into a category.
- Model coverage, known limitations, error patterns and intended use will be documented.

14. Expected Outcomes and Deliverables

- A documented pilot dataset and complaint annotation guidelines.
- A reproducible TF-IDF classification baseline.
- A comparison with a multilingual transformer model, if compute and data permit.
- An urgency suggestion module with visible evidence cues and an abstention option.
- A semantic duplicate-retrieval module evaluated on labelled complaint pairs.
- A Streamlit demonstration using synthetic or authorised examples.
- An evaluation report covering per-language performance, limitations and error analysis.

15. Limitations and Future Scope

The initial prototype will not cover every Indian language, dialect or local spelling convention. Small or synthetic datasets may not represent real reporting patterns. Text-based location extraction cannot verify a physical location, and semantic similarity cannot prove that two reports refer to one incident. Urgency labels may also be subjective without domain-expert guidance.

Future work may include additional Indian languages, speech-to-text intake, temporal and geographic incident aggregation, active learning, calibrated confidence estimates and integration with an authorised civic-service workflow after security and governance review.

16. Conclusion

CivicSift AI proposes a feasible and socially relevant application of multilingual NLP. By combining complaint classification, evidence-aware urgency suggestions and semantic duplicate discovery, the project aims to help reviewers organise unstructured reports and identify related issues. Its research value will depend on transparent baselines, careful language-specific evaluation, responsible data handling and honest reporting of limitations. A six-week prototype can demonstrate the core workflow without requiring training a large language model from scratch.

17. References

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- Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. Proceedings of EMNLP-IJCNLP 2019. https://aclanthology.org/D19-1410/
- Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825–2830. https://jmlr.org/papers/v12/pedregosa11a.html

Appendix A. Illustrative Examples

Illustrative complaint	Possible system response (not measured)
“There is water leakage near the school since morning.”	Water category; urgency to be assessed from agreed guidelines; show “leakage” and “since morning” as text cues.
“Street light not working for 3 days.”	Electricity / street-lighting category; show duration and failure phrase.

“Road pe bada hole hai, bikes are falling.”	Roads category; code-mixed text flag; possible high-priority review cue “bikes are falling”.

The examples above are synthetic illustrations. They are not real citizen reports, validated labels or model predictions.

Appendix B. Proposed Project Folder Structure

```
civicsift-ai/
├── data/ # approved data only
├── notebooks/ # exploration and experiments
├── src/
│ ├── preprocessing.py
│ ├── classify.py
│ ├── urgency.py
│ ├── duplicate_search.py
│ └── evaluation.py
├── app/
│ └── streamlit_app.py
├── reports/
├── requirements.txt
└── README.md
```

Appendix C. Final Submission Checklist

- Fill in the student, internship, mentor and organisation details on the cover.
- Confirm the project title and language scope with the internship supervisor.
- Verify the latest bibliographic details and dataset/model licences before implementation.
- Keep all target metrics labelled as proposed targets until experiments are completed.
- Use only authorized or synthetic data in screenshots and demonstrations.
- Run a final spelling and formatting check before submission.